

Neural Network

1. Component of a neural network and discuss the similarities of neural network.

A neural network is a computation model inspired by human brain. And its component are: (i) Input layer (ii) Hidden layers

(iii) Output layer (iv) Neuron Node (v) weights (vi) Bias (vii) Activation function (viii) Loss function (ix) Optimizer

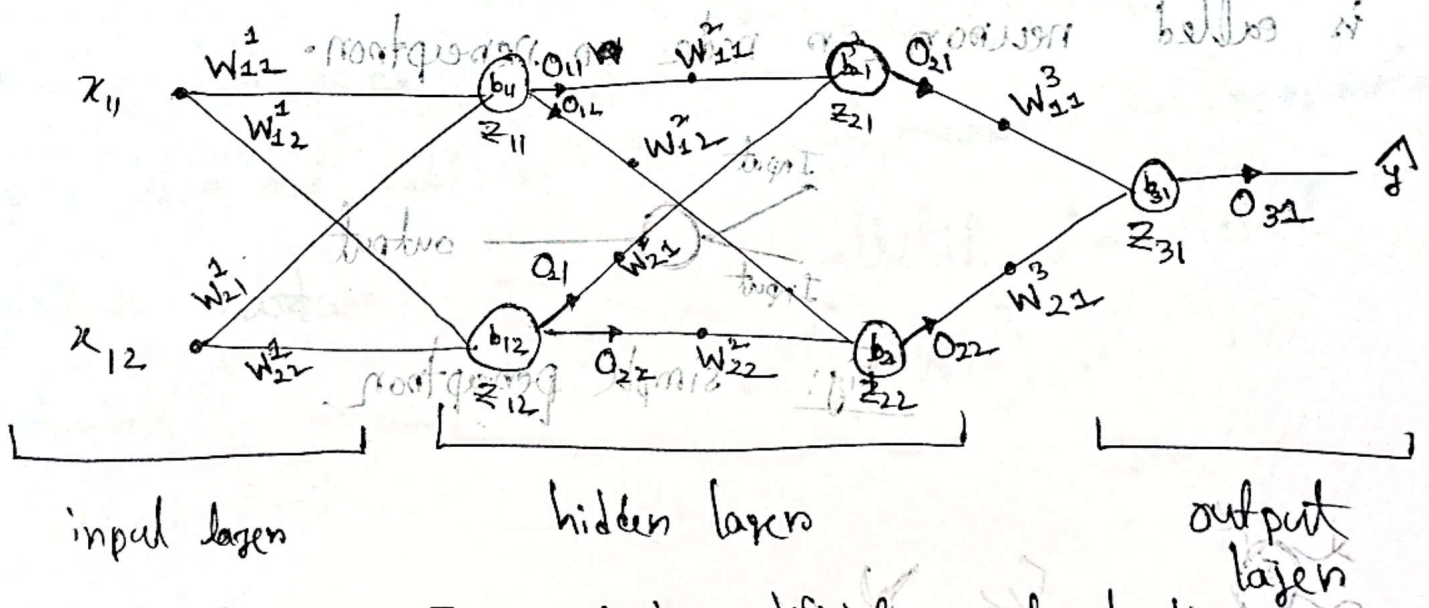


Fig: simple artificial neural network

(Loss $\rightarrow y \rightarrow z_{31} \rightarrow O_{21} \rightarrow z_{11} \rightarrow O_{11} \rightarrow z_{11} \rightarrow W_{11}$)

Computational graph for updating W_{11}

$$\frac{\partial L}{\partial W_{11}} = \frac{\partial L}{\partial y} \times \frac{\partial y}{\partial z_{31}} \times \frac{\partial z_{31}}{\partial O_{21}} \times \frac{\partial O_{21}}{\partial z_{11}} \times \frac{\partial z_{11}}{\partial O_{11}} \times \frac{\partial O_{11}}{\partial z_{11}} \times \frac{\partial z_{11}}{\partial W_{11}}$$

Weighted Input

$$Z_{11} = X_{11}W_{11}^1 + X_{21}W_{21}^1 + b_{11}$$

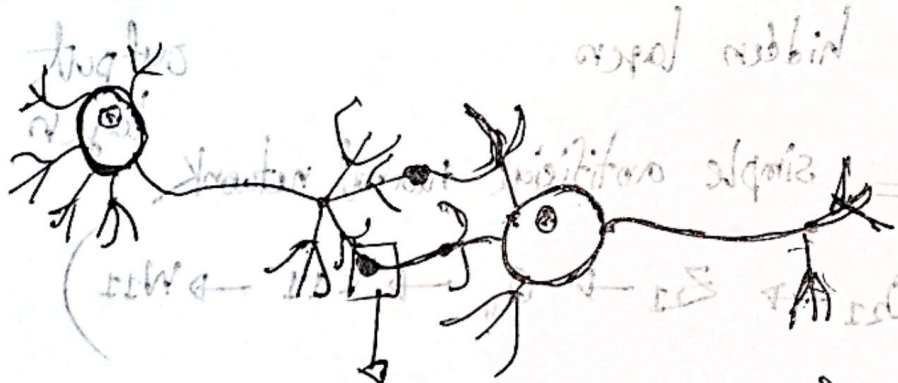
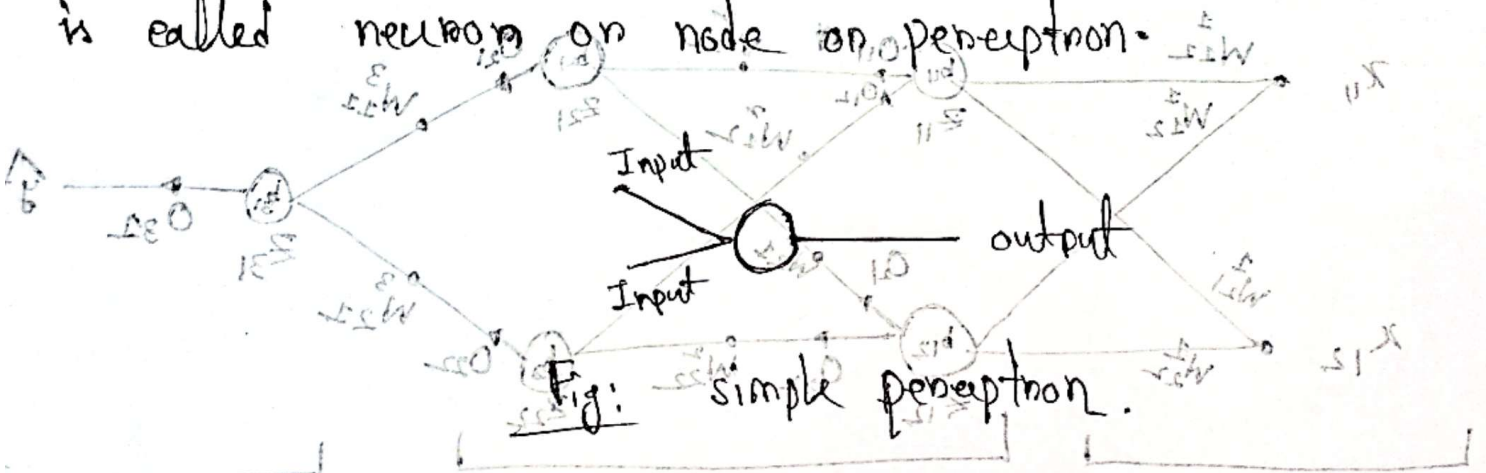
$O_{11} = \text{Activation function}(Z_{11})$

Known known to

$Z_{21} = O_{11}W_{12}^2 + O_{21}W_{22}^2 + b_{21}$

$O_{21} = \text{Activation function}(Z_{21})$

here, the ANN made up multiple neuron on hidden layers
 call per. The smallest unit of a neural network
 is called neuron or node or perception.



Neuron 1

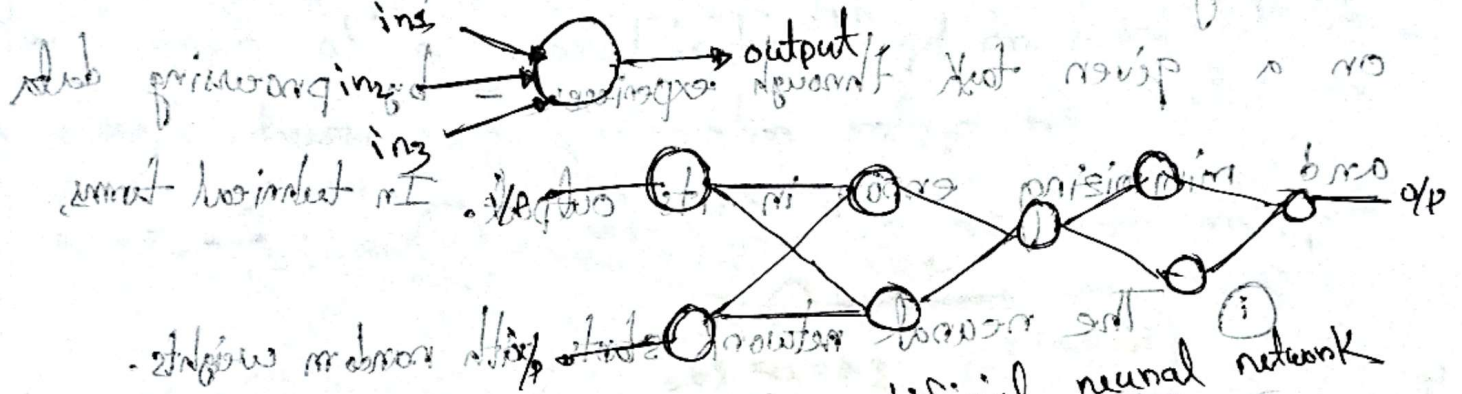
Synapse

Neuron 2

A human brain has billion of neurons. All the neur

interconnected with each other making a big network called neural network. Similarly, a perception on

it's called artificial neural network.



Similar, thousand of artificial neurons (number based on input data and model complexity) interconnected with each other

make a network and we call it Artificial Neural Network (ANN).

Using the backpropagation and gradient descent

the network adjusts its weights

Reporting the process called training.

The process of giving input and output to the network is called training.

What does "learning" mean in the context of neural networks?

Learning means that a neural network improves its performance on a given task through experience by processing data and minimizing error in its output. In technical terms,

(i) The neural network starts with random weights.

(ii) It processes the data through its layers (called forward propagation).

(iii) The output is compared with the expected output to find the error.

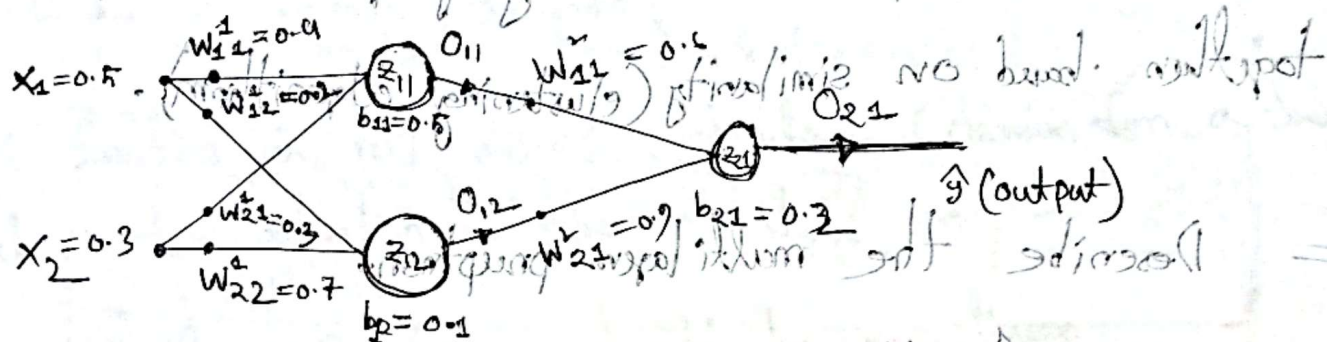
(iv) The difference between them is error or loss.

(v) Using the backpropagation and gradient descent, the neural network adjusts its weights.

(vi) Repeating this process many times called (training).

Q Explain forward propagation and self-organization.

Forward propagation is the process of moving inputs through the layers of a neural network to get an output. Let's take an example below.



Dataset

x_1	x_2	Target
0.5	0.3	1

and, let's take activation function as sigmoid.

$$o_{11} = \frac{1}{1 + e^{-z_{11}}} = \frac{1}{1 + e^{-(0.5 \times 0.4 + 0.3 \times 0.2 + 0.5)}} = 0.68$$

$$o_{12} = \frac{1}{1 + e^{-z_{12}}} = \frac{1}{1 + e^{-(0.5 \times 0.2 + 0.3 \times 0.7 + 0.1)}} = 0.09$$

$$o_{21} = \frac{1}{1 + e^{-z_{21}}} = \frac{1}{1 + e^{-(0.68 \times 0.6 + 0.09 \times 0.9 + 0.2)}} = 0.76$$

$$\hat{y} = o_{21} = 0.76$$

The process of getting output from input is called forward propagation.

Self-Organization Self organization is a learning

process without supervision (unsupervised learning). The network organizes data itself and finds a pattern of data. The system automatically groups similar inputs together based on similarity (clustering algorithm).

40 Describe the multi-layer perceptron

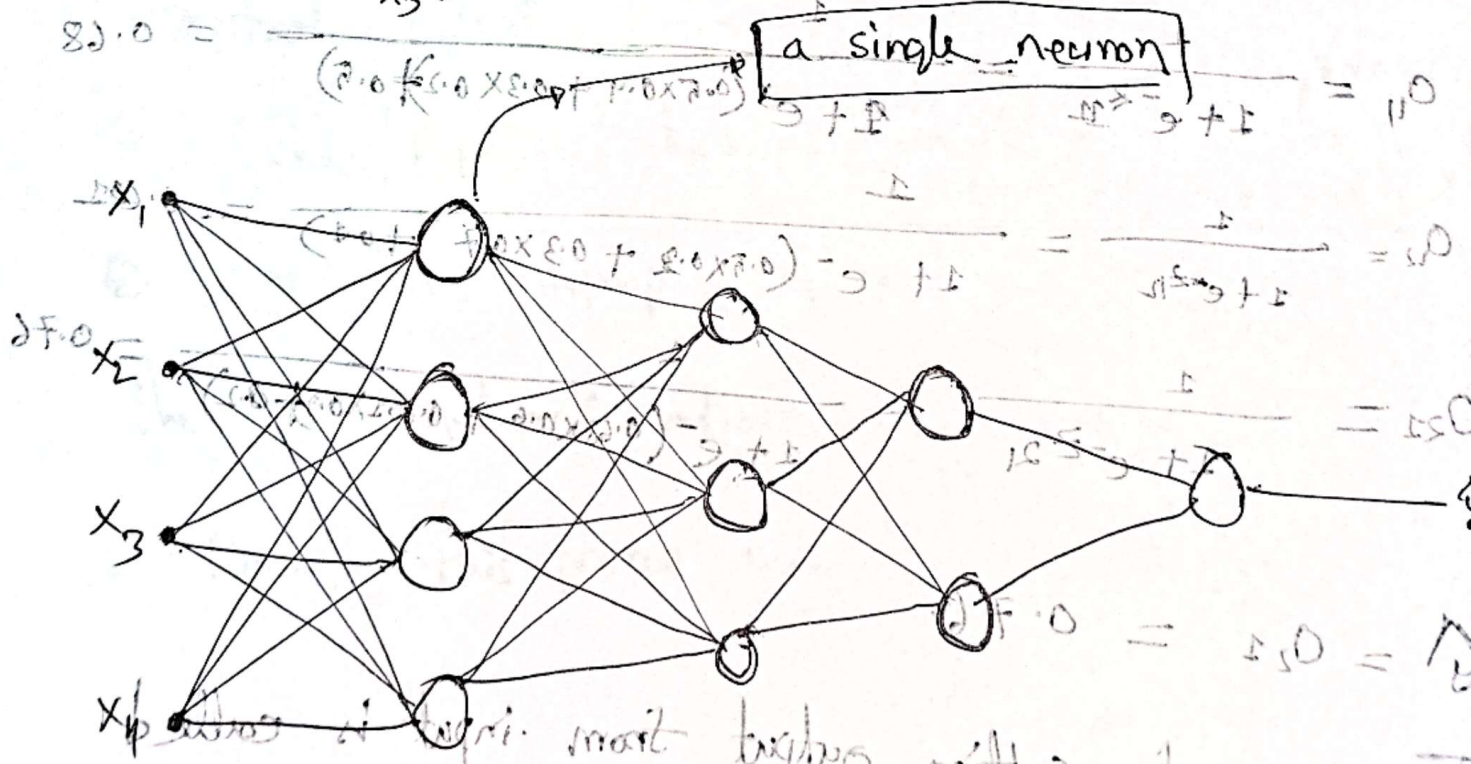
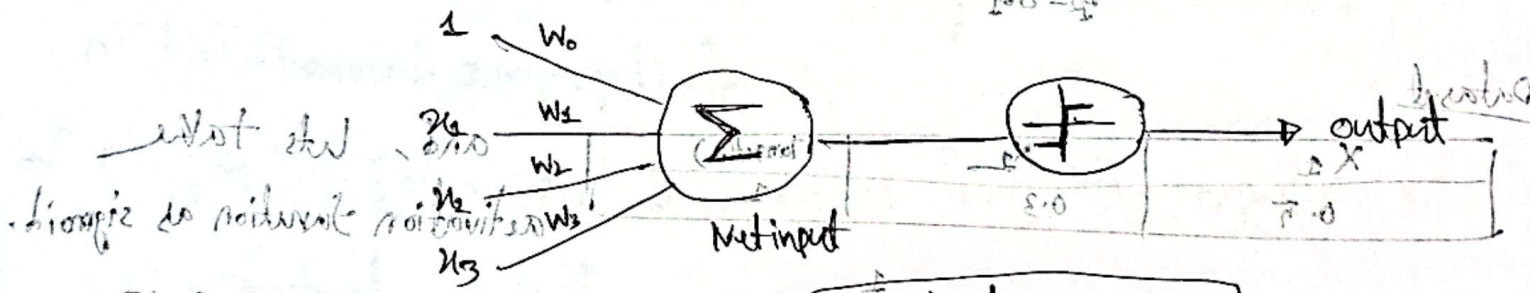


Fig: Multi-layer Perceptron (MLP)

5. Illustrate and describe the standard activation function.

In artificial neural network each weighted sum of input passes through a scalar value function called activation function.

All function is not belong to activation on step.)

Criteria for a function

- (i) Activation function should be non-linear.
- (ii) Activation function should be differentiable.
- (iii) Activation function should be numerically inexpensive.
- (iv) Activation function should be zero centred.
- (v) Activation function should be non-saturating.

means bound within a range. Like, saturating function sigmoid output range $(0 \sim 1)$ and tanh output range $(-1, 1)$. For very very higher input sigmoid give always 1 same for tanh that why it's faces vanishing gradient problem.

Activation function with formula

Function	Formula	Range	Remarks
Sigmoid	$f(x) = \frac{1}{1+e^{-x}}$	$(0, 1)$	prone to vanishing gradient
Relu	$f(x) = \max(0, x)$	$(0, \infty)$	may cause "dead neuron" when input $x \leq 0$
Leaky Relu	$f(x) = \begin{cases} x, & x > 0 \\ 0.01x, & x \leq 0 \end{cases}$	$(-\infty, \infty)$	Solve the "dead neuron" problem
Tanh	$f(x) = \frac{e^x - e^{-x}}{e^x + e^{-x}}$	$(-1, 1)$	zero-centered (normalize) fast convergence than sigmoid
softmax	$f(x) = \frac{e^{x_i}}{\sum_j e^{x_j}}$	$(0, 1)$	Converts output into probability

probabilities = norm

probabilities like $\frac{1}{1+e^{-x}}$ for output

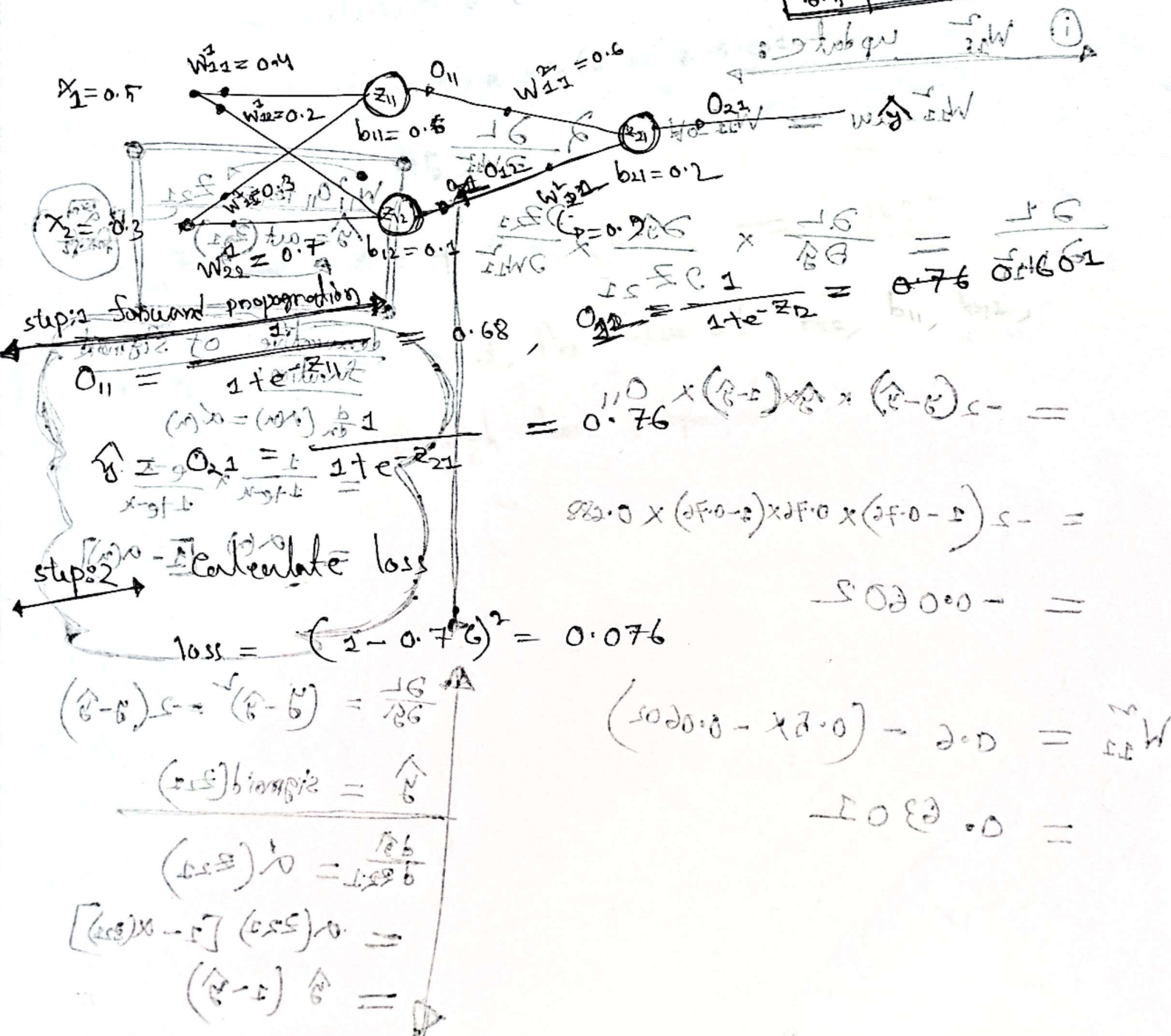
probabilities like $\max(0, x)$ for output

probabilities like $\frac{e^x - e^{-x}}{e^x + e^{-x}}$ for output

probabilities like $\frac{e^{x_i}}{\sum_j e^{x_j}}$ for output

6. How to adjust the weights of perceptron in mlp?
 on state the back-propagation algorithm for learning in a multilayer network.

Let's take the below mlp



Ex. We have to design soft layer of wall

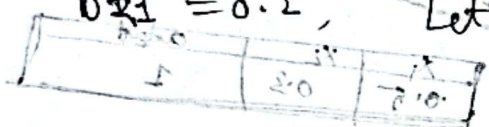
$x_{11} = 0.5$ $x_{12} = 0.3$ $y = 1$ (target)

$O_{11} = 0.688$ $O_{12} = 0.601$, $\hat{y} = O_{21} = 0.76$

$W_{11}^2 = 0.6$, $W_{12}^2 = 0.9$

$b_{11} = 0.2$

Let's $\lambda = 0.5$ for weight update



① W_{11}^2 update:

$$W_{11}^2_{\text{new}} = W_{11}^2_{\text{old}} - \lambda \frac{\partial L}{\partial W_{11}^2}$$

$$\frac{\partial L}{\partial W_{11}^2} = \frac{\partial L}{\partial \hat{y}} \times \frac{\partial \hat{y}}{\partial z_{21}} \times \frac{\partial z_{21}}{\partial W_{11}^2}$$

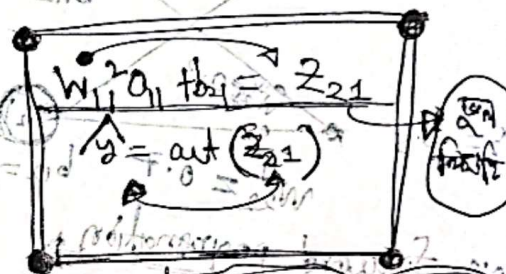
$$= -2(y - \hat{y}) \times \hat{y}(1 - \hat{y}) \times O_{11}$$

$$= -2(1 - 0.76) \times 0.76 \times (1 - 0.76) \times 0.688$$

$$= -0.0602$$

$$W_{11}^2 = 0.6 - (0.5 \times -0.0602)$$

$$= 0.6301$$



derivative of sigmoid function:

$$\frac{d}{dx} \sigma(x) = \sigma'(x)$$

$$= \frac{1}{1 + e^x} \times \frac{e^{-x}}{1 + e^{-x}}$$

$$= \sigma(x) [1 - \sigma(x)]$$

$$\frac{\partial L}{\partial \hat{y}} = (y - \hat{y})^2 = -2(y - \hat{y})$$

$$\hat{y} = \text{sigmoid}(z_{21})$$

$$\frac{\partial \hat{y}}{\partial z_{21}} = \sigma'(z_{21})$$

$$= \sigma(z_{21}) [1 - \sigma(z_{21})]$$

$$= \hat{y} (1 - \hat{y})$$

(ii) For, W_{21}^1 ,

$$Z_{21} = 0_{11} W_{11}^2 + 0_{12} W_{21}^2 + b_{21} \quad \text{--- (i)}$$

$$\hat{y} = \text{sigmoid}(Z_{21}) \quad \text{--- (ii)}$$

$$\frac{\partial L}{\partial W_{21}^2} = \frac{\partial L}{\partial \hat{y}} \times \frac{\partial \hat{y}}{\partial Z_{21}} \times \frac{\partial Z_{21}}{\partial W_{21}^2}$$

$$= -2(y - \hat{y}) \times \hat{y} \times (1 - \hat{y}) \times 0_{12}$$

$$= -2(1 - 0.76) \times 0.76 (1 - 0.76) \times 0.601$$

$$= -0.053$$

$$W_{12}^2 = 0.9 + (0.5)(0.53) = 1.165$$

Similarly, we will get the value of b_{21} , b_{11} , b_{12} , W_{11}^1 , W_{12}^1 through back propagation.